

JANMESH JOSHI

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[Github.com](#) | [Portfolio](#) | [LinkedIn/janmeshjoshi](#)

ML Engineer with 4+ years of experience building production machine learning systems, semantic search, RAG pipelines, and scalable inference services. Improved top-5 retrieval relevance by 35% across 500K+ legal documents and deployed containerized ML services serving 1,000+ daily requests with sub-100ms latency. MSc in Human-Centered AI (TU Dublin) specializing in Explainable AI and AI Governance for regulated environments.

TECHNICAL SKILLS

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, RAG, Hugging Face, Langchain
LLMs/NLP: GPT-4o, Claude, BERT, BERTopic, UMAP, DPR, BGE-M3, Sentence-Transformers, FAISS, HNSW
MLOps, Cloud & Monitoring: AWS, Docker, Google Cloud Run, CI/CD, Git, Kubernetes
API & Backend: FastAPI, REST APIs, Python, SQL, PostgreSQL

EXPERIENCE

Founder & Product Engineer <i>MyNextRoom</i> Link	January 2026 - Present <i>Dublin, Ireland</i>
- Sole founder and engineer of a cross-platform rental marketplace built using React Native/Expo, PostgreSQL, FastAPI, and Google Cloud Run, featuring real-time messaging, push notifications, universal deep linking, and AI-powered fraud detection. - Trained and deployed a Random Forest fraud detection classifier on listing price-to-area feature vectors, achieving automated anomaly scoring for real-time listing moderation served via FastAPI on Google Cloud Run. - Built a k-NN compatibility matching system using Scikit-learn on structured renter-listing feature vectors spanning budget, move-in date, and housing preferences, replacing rule-based filtering with a learned similarity model.	
Machine Learning Engineer <i>Panghat</i> , Legal-tech startup building AI search over case law	June 2020 - Aug. 2024 <i>Mumbai, India</i>
- Improved top-5 semantic retrieval relevance by 35% over BM25 baseline across 500K+ legal documents using DPR-CTX and BGE embeddings. - Built and deployed containerized ML inference services on Docker and Kubernetes, serving 1000+ daily requests with sub-100ms latency. - Reduced vector search latency by 10x by HNSW index tuning and optimized partitioning strategies in Milvus. - Developed a RAG-based legal search assistant using LangChain and GPT-4, achieving 92% answer relevance in internal evaluations.	
Teaching Assistant (TA) <i>Technological University of Dublin</i>	Feb. 2025 - Present <i>Dublin, Ireland</i>
Delivered Machine Learning and Python labs to 50+ undergraduate students, covering supervised learning, neural networks, data structures, and clean software engineering practices while conducting code reviews and mentoring students.	

PROJECTS

Ethical AI & Risk Modeling for AI Tutors (MSc Thesis) <i>Technological University of Dublin</i>	Sept 2024 - Oct 2025 <i>Dublin, Ireland</i>
- Engineered a Machine Learning pipeline to detect user over-reliance on AI systems, achieving a Macro F1 score of 0.89 using Random Forest Classifiers. - Implemented Explainable AI (SHAP) to interpret model decisions, identifying behavioural patterns associated with AI over-reliance and supporting transparent model evaluation.	

EDUCATION

Technological University of Dublin <i>Master of Science, Human Centered Artificial Intelligence</i>	Sept. 2024 - Oct. 2025 <i>Dublin, Ireland</i>
University of Mumbai <i>Master of Science, Information Technology</i>	Aug. 2020 - May 2022 <i>Mumbai, India</i>
University of Mumbai <i>Bachelor of Science, Information Technology</i>	Aug. 2017 - May 2020 <i>Mumbai, India</i>